

Problem Statements

Domain 1: EMBEDDED SYSTEMS & IOT

Problem 1 – Smart Motor Condition Monitoring

Design and develop an embedded system to monitor the temperature, vibration and operating status of a small motor. The system should identify abnormal conditions and provide a visual/audio warning.

Problem 2 – IoT-Based Machine Utilization Monitor

Develop an IoT-enabled system to monitor **machine operating time, idle time and production cycles** and calculate the approximate machine utilization.

The system should transmit the information to a local IoT dashboard.

Problem 3 – Smart Industrial Environment Monitoring

Develop an IoT-based monitoring system for an industrial workspace that measures temperature, humidity and air-quality/gas indication and generates an alert when predefined limits are exceeded.

Problem 4 – Connected Worker Safety Alert

Develop a wearable/portable embedded device capable of detecting a **fall or abnormal inactivity** and generating an emergency alert.

An additional sensor may be used to demonstrate restricted-area entry detection.

Domain 2: INDUSTRIAL AUTOMATION & PLC

Problem 5 – PLC-controlled Traffic control system with automatic time adjustment according to traffic density.

Problem 6 – PLC-controlled conveyor with automatic rejection & Energy saving

Problem 7 – PLC-controlled automatic parking space signalling system.

Problem 8 – PLC-based tank level control with 2 pumps in Duty + Assist configuration.

Problem 9 – PLC-based OEE (Overall Equipment Effectiveness) program.

Problem 10 – PLC-Based Preventive Maintenance (PM) Warning System

weekly - Lubrication ,cleaning

Monthly - Hydraulic Oil Check, Filter Change

Quarterly - Pneumatics Check.

Yearly - Sensor Calibration

Domain 3: ROBOTICS & AUTOMATION

Problem 9 – Vision-Based Object Classification and Sorting

Develop a camera-based system that identifies objects according to colour, shape or category and activates an actuator to direct them to the appropriate location.

Problem 10 – Obstacle-Avoiding Mobile Robot

Develop a mobile robot that moves along a predefined path and detects obstacles placed in its path.

When an obstacle is detected, the robot should stop, change direction and continue its movement.

Problem 11 – Autonomous Station-to-Station Delivery Robot

Develop a small mobile robot capable of transporting a lightweight object between two or three predefined stations.

Problem 12 – Automated Pick-and-Place Prototype

Develop a simple robotic pick-and-place system that detects an object at a predefined pickup location and transfers it to a predefined placement location.

Domain 4: INDUSTRIAL COMMUNICATION & INDUSTRY 4.0

Problem 13 – MQTT-Based Industrial Machine Dashboard

Develop an IIoT system in which a machine/controller sends parameters such as temperature, machine status, production count and fault status to a central dashboard using MQTT.

Problem 14 – Wireless Multi-Node Industrial Monitoring

Develop a wireless monitoring system consisting of two or three sensor nodes that transmit industrial parameters to a central gateway.

Problem 15 – Mini Smart Factory Data Monitoring System

Develop a miniature Industry 4.0 model consisting of two simulated machines connected to a common monitoring platform.

Domain 5: AI & SMART AUTOMATION

Problem 16 – AI/Computer-Vision-Based Quality Inspection

Develop a camera-based inspection system that classifies a product as: **GOOD / DEFECTIVE based on one or more predefined characteristics such as colour, shape, size or missing**

Problem 17 – Sensor-Based Machine Anomaly Detection

Develop a system that monitors temperature, vibration or current and identifies whether the machine is operating under normal or abnormal conditions.

Problem 18 – Intelligent Production Counter

Develop an intelligent production-monitoring system that detects products passing through a production station and identifies normal and abnormal production conditions.

Domain 6: ENERGY & SUSTAINABLE ELECTRONICS

Problem 19 – Smart Priority-Based Energy Management

Develop an intelligent load-management system that controls multiple loads according to their priority when the available power is limited.

Problem 20 – Solar-Battery Intelligent Load Controller

Develop a low-voltage energy-management system that intelligently selects the available energy source for supplying a load.